

From a Stork to a Squid:
Lessons Learned from
Migrating MinIO to Ceph

Our Heroes



MinIO is a high-performance, **S3** compatible **object** store, open sourced under GNU AGPLv3 license.

Rook is an open-source cloud-native storage **orchestrator** for Kubernetes, providing the platform, framework, and support for Ceph storage to natively integrate with Kubernetes.

Ceph is a free and open-source software-defined storage platform that provides **object** storage, **block** storage, and **file storage** built on a common distributed cluster foundation

Why change?

- DORA came into play in 17th of January 2025
- Many risk and compliance reviews start happening

22-01 13:44
Spoke with Per about this yesterday - we need to get out of this
Using anything with AGPL-3.0



Component Architecture Differences

----- Internode Traffic		>minioClient.getObject()						
NAMESPACE↑	NAME	PF	READY	STATUS	RESTARTS	CPU	MEM	%CP
minio-applications	minio-tenant-applications-pool-0-0	●	3/3	Running	0	9	363	
minio-applications	minio-tenant-applications-pool-0-1	●	3/3	Running	0	10	320	1
minio-applications	minio-tenant-applications-pool-0-2	●	3/3	Running	0	11	299	1
minio-applications	minio-tenant-applications-pool-0-3	●	3/3	Running	0	14	129	1
minio-applications	minio-tenant-applications-pool-0-4	●	3/3	Running	0	10	275	1
minio-applications	minio-tenant-applications-pool-0-5	●	3/3	Running	0	10	242	1
minio-applicationsci	minio-tenant-applicationsci-pool-0-0	●	3/3	Running	0	17	340	1
minio-applicationsci	minio-tenant-applicationsci-pool-0-1	●	3/3	Running	0	46	175	4
minio-applicationsci	minio-tenant-applicationsci-pool-0-2	●	3/3	Running	0	15	365	1
minio-applicationsci	minio-tenant-applicationsci-pool-0-3	●	3/3	Running	0	23	289	2
minio-applicationsci	minio-tenant-applicationsci-pool-0-4	●	3/3	Running	0	16	304	1
minio-applicationsci	minio-tenant-applicationsci-pool-0-5	●	3/3	Running	0	23	314	2
minio-platform	minio-tenant-platform-pool-0-0	●	3/3	Running	0	9	270	
minio-platform	minio-tenant-platform-pool-0-1	●	3/3	Running	0	8	309	
minio-platform	minio-tenant-platform-pool-0-2	●	3/3	Running	0	13	311	1
minio-platform	minio-tenant-platform-pool-0-3	●	3/3	Running	0	9	385	
minio-platform	minio-tenant-platform-pool-0-4	●	3/3	Running	0	9	278	
minio-platform	minio-tenant-platform-pool-0-5	●	3/3	Running	0	8	310	
minio-platform	minio-tenant-platform-pool-1-0	●	3/3	Running	0	12	256	1
minio-platform	minio-tenant-platform-pool-1-1	●	3/3	Running	0	7	248	
minio-platform	minio-tenant-platform-pool-1-2	●	3/3	Running	0	8	281	
minio-platform	minio-tenant-platform-pool-1-3	●	3/3	Running	0	7	301	
minio-platform	minio-tenant-platform-pool-1-4	●	3/3	Running	0	8	271	
minio-platform	minio-tenant-platform-pool-1-5	●	3/3	Running	0	21	135	2
operators	iitech-minio-client-operator-845fd496b6-wbp8m	●	2/2	Running	0	2	36	
operators	minio-operator-97f44d6c7-jd7l6	●	2/2	Running	0	12	66	
operators	minio-operator-97f44d6c7-vjlfq	●	2/2	Running	0	2	46	

NAMESPACE↑	NAME	PF	READY	STATUS	RESTARTS	CPU	MEM
rook-ceph	csi-cephfsplugin-5q76d	●	3/3	Running	2	1	35
rook-ceph	csi-cephfsplugin-7lknc	●	3/3	Running	0	1	36
rook-ceph	csi-cephfsplugin-46sxl	●	3/3	Running	7	1	44
rook-ceph	csi-cephfsplugin-bm4f8	●	3/3	Running	7	1	39
rook-ceph	csi-cephfsplugin-f5qcs	●	3/3	Running	7	1	32
rook-ceph	csi-cephfsplugin-fzd2t	●	3/3	Running	2	1	29
rook-ceph	csi-cephfsplugin-gbfm5	●	3/3	Running	2	1	28
rook-ceph	csi-cephfsplugin-gpqmz	●	3/3	Running	2	1	38
rook-ceph	csi-cephfsplugin-j4dz5	●	3/3	Running	0	1	33
rook-ceph	csi-cephfsplugin-k4b2m	●	3/3	Running	7	1	36
rook-ceph	csi-cephfsplugin-k7x2q	●	3/3	Running	10	1	32
rook-ceph	csi-cephfsplugin-lcg2t	●	3/3	Running	12	1	42
rook-ceph	csi-cephfsplugin-m8h9w	●	3/3	Running	2	1	36
rook-ceph	csi-cephfsplugin-mlmnf	●	3/3	Running	10	1	28
rook-ceph	csi-cephfsplugin-ncc7n	●	3/3	Running	7	1	38
rook-ceph	csi-cephfsplugin-ncwgf	●	3/3	Running	2	1	37
rook-ceph	csi-cephfsplugin-p4rjn	●	3/3	Running	5	1	36
rook-ceph	csi-cephfsplugin-phwt7	●	3/3	Running	0	1	34
rook-ceph	csi-cephfsplugin-provisioner-989bff6f4-5vfmq	●	6/6	Running	0	3	93
rook-ceph	csi-cephfsplugin-provisioner-989bff6f4-kv2g8	●	6/6	Running	0	2	68
rook-ceph	csi-cephfsplugin-vb4ps	●	3/3	Running	2	1	34
rook-ceph	csi-cephfsplugin-vjsmt	●	3/3	Running	10	1	36
rook-ceph	csi-cephfsplugin-xj8h5	●	3/3	Running	10	1	28
rook-ceph	csi-cephfsplugin-z5gzd	●	3/3	Running	10	1	38
rook-ceph	csi-cephfsplugin-zltgk	●	3/3	Running	10	1	29
rook-ceph	csi-rbdplugin-2g7pj	●	3/3	Running	5	1	43
rook-ceph	csi-rbdplugin-5dzwf	●	3/3	Running	10	1	39
rook-ceph	csi-rbdplugin-5mmvs	●	3/3	Running	7	1	42
rook-ceph	csi-rbdplugin-6jpm	●	3/3	Running	10	1	33
rook-ceph	csi-rbdplugin-8gq2g	●	3/3	Running	0	1	46
rook-ceph	csi-rbdplugin-8krll	●	3/3	Running	7	1	36
rook-ceph	csi-rbdplugin-88kz8	●	3/3	Running	12	1	49
rook-ceph	csi-rbdplugin-cht47	●	3/3	Running	2	1	43
rook-ceph	csi-rbdplugin-cjgsv	●	3/3	Running	10	1	36
rook-ceph	csi-rbdplugin-dv5lf	●	3/3	Running	2	1	39
rook-ceph	csi-rbdplugin-fr6vf	●	3/3	Running	0	1	37
rook-ceph	csi-rbdplugin-h6vzd	●	3/3	Running	2	1	34
rook-ceph	csi-rbdplugin-jpzvz	●	3/3	Running	7	1	56
rook-ceph	csi-rbdplugin-mtsz9	●	3/3	Running	7	1	39
rook-ceph	csi-rbdplugin-p4g7h	●	3/3	Running	2	1	46

Developers Hate Migrations

- Bad timing
 - Migration was announced shortly after we advertised that our MinIO platform was ready and before we had a clear path to Rook Ceph
- Need to make it painless for developers
 - We decided to create a generic crossplane provider for object storages that would allow simple changes in the future
- Need to make it easy for the future
 - With this we have 1 CRD with which we can easily swap between object storage backends, by only changing the apiVersion number

Karlis 15:11

Hey @everyone, remember when we encouraged you all to use our MinIO platform?

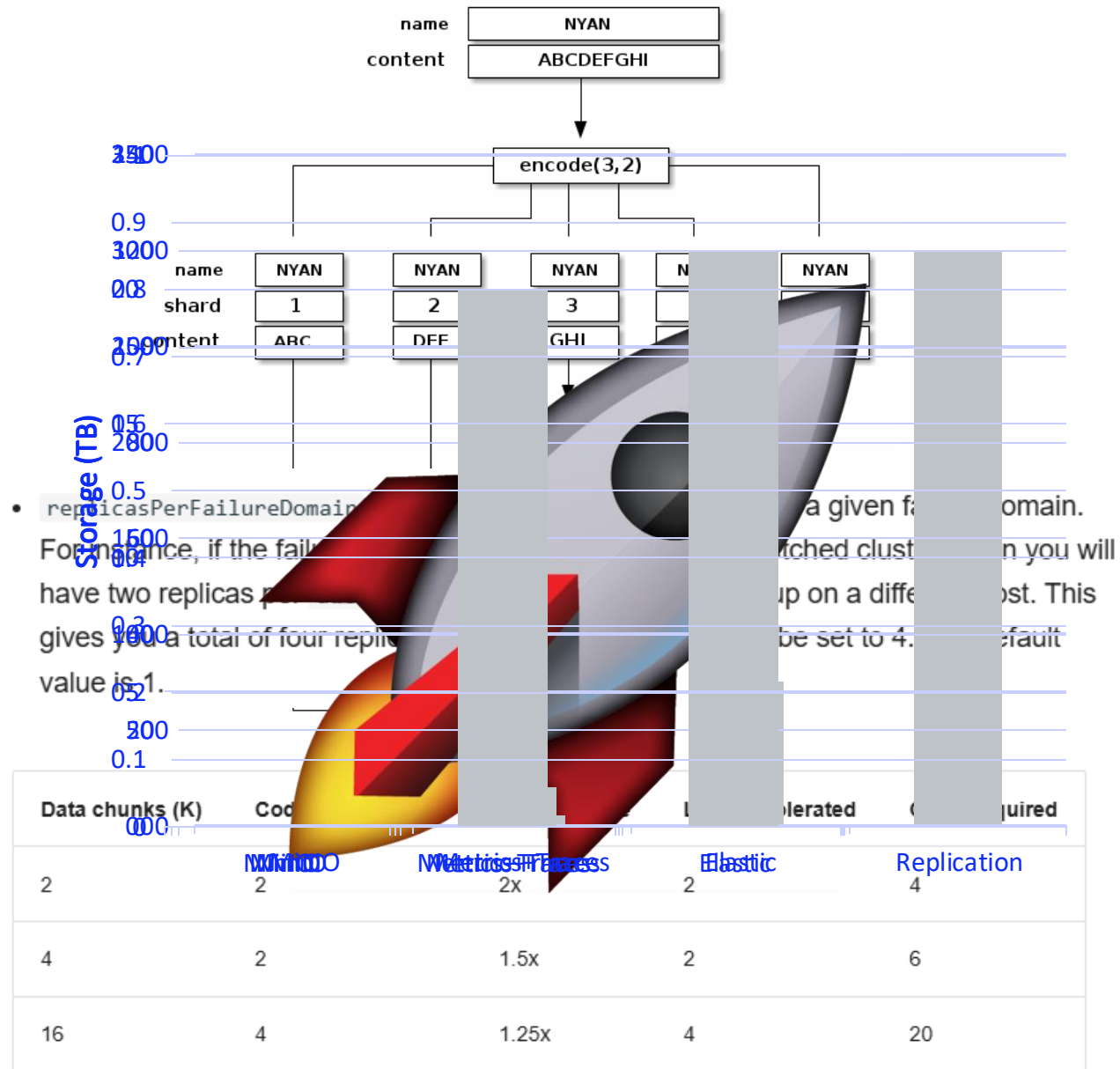
```
---
apiVersion: minioclient.iitech/v1
kind: MinioBucket
metadata:
  name: mimir-alertmanager-storage
  namespace: monitoring
spec:
  tenantName: platform
  fullName: mimir-alertmanager-storage
  onDeletePolicy: ignore
---
apiVersion: minioclient.iitech/v1
kind: MinioUser
metadata:
  name: mimir-minio-user
  namespace: monitoring
spec:
  tenantName: platform
  userName: mimir-minio-user
  secretKeySecret: mimir-minio-secret
---
apiVersion: minioclient.iitech/v1
kind: MinioPolicy
metadata:
  name: mimir-minio-policy-all
  namespace: monitoring
spec:
  tenantName: platform
  policyName: mimir-minio-policy-all
  # version: "2012-10-17"
statements:
  - effect: "Allow"
    actions: [ "s3:GetObject", "s3:PutObject", "s3:DeleteObject", "s3:ListBucket" ]
    resources: [ "arn:aws:s3::mimir-blocks-storage/*" ]
  - effect: "Allow"
    actions: [ "s3:GetObject", "s3:PutObject", "s3:DeleteObject", "s3:ListBucket" ]
    resources: [ "arn:aws:s3::mimir-alertmanager-storage/*" ]
  - effect: "Allow"
    actions: [ "s3:GetObject", "s3:PutObject", "s3:DeleteObject", "s3:ListBucket" ]
    resources: [ "arn:aws:s3::mimir-ruler-storage/*" ]
  users:
  - mimir-minio-user
---
apiVersion: iitech.dk/v1alpha1
kind: Bucket
metadata:
  name: mimir-alertmanager-storage
  namespace: monitoring
spec:
  deletionPolicy: Orphan
  forProvider:
    name: mimir-alertmanager-storage
    owner:
      id: mimir-ceph-user
    policy:
      structured:
        statements:
          - actions:
              - s3:GetObject
              - s3:PutObject
              - s3:DeleteObject
              - s3:ListBucket
            effect: Allow
            resources:
              - mimir-alertmanager-storage/
            principals:
              - mimir-ceph-user
        version: "2012-10-17"
    providerConfigRef:
      name: platform-object-storage
---
apiVersion: iitech.dk/v1alpha1
kind: User
metadata:
  name: mimir-ceph-user
  namespace: monitoring
spec:
  forProvider:
    name: mimir-ceph-user
    providerConfigRef:
      name: platform-object-storage
```

Developer2 15:13

WHAT?!

Migration and Expansion

- Most teams have migrated with ease, thanks to the crossplane provider
- We were comfortably storing ~40GB until now in MinIO
- But the usage grew heavily:
 - Our metrics and traces are now stored there (~20TB)
 - Elastic team wants to use it too (~300TB)
- Now we need to store >1PB
 - Due to us having running Rook Ceph in stretched cluster mode we can't use erasure coding – so we can't have 1.5x replication
 - We are forced into 4x replication



Lessons Learned

- Avoid making hard references to specific vendors/products in your platform – be more generic
 - Instead of Redis, use Cache
 - Instead of MinIO, use ObjectStorage
 - Instead of Linkerd, use Mesh
 - Instead of Kyverno, use Compliance/Policies
- Utilize tools like Crossplane and Kyverno to improve migrations and developer experience
- Licenses are there for a reason – keep an eye on them, as they tend to change

One year ago Redis changed its license – and lost most of its external contributors

By Tim Anderson - April 1, 2025

Synadia tries to “withdraw” the NATS project from the CNCF and relicense to BSL non-open source license

Protecting NATS and the integrity of open source: CNCF’s commitment to the community



Thank you for your
attention!

